Question 7

1 Introduction

The objective of this project was to predict the risk of diabetes using a patient's clinical features. We implemented and compared two distinct machine learning models: an ANFIS-like fuzzy inference system and a Multilayer Perceptron (MLP). The dataset used for this task was the Pima Indians Diabetes Dataset, which includes features such as BMI, age, glucose levels, and blood pressure, with the 'Outcome' column as the target variable (0 for healthy, 1 for diabetic). A key focus of this project was to evaluate not only the accuracy of each model but also its **interpretability**, which is crucial in a medical context where a doctor needs to understand and explain a diagnosis.

2 Model Implementation and Analysis

2.1 ANFIS-like Fuzzy System

An ANFIS (Adaptive Neuro-Fuzzy Inference System) combines the human-like reasoning of a fuzzy system with the learning capabilities of a neural network. However, since a dedicated ANFIS tool is not readily available in Python, we created a similar fuzzy inference system using the `scikit-fuzzy` library. This model was designed to be highly interpretable.

- **Inputs:** To avoid the "curse of dimensionality," the model was trained using only two key features: **Normalized BMI** and **Normalized Age**.
- **Methodology:** The data was first normalized to a range of $[0, 1]$ to ensure all values fell within the defined fuzzy sets. We then defined fuzzy membership functions (e.g., 'low', 'medium', 'high' for BMI and 'young', 'middle', 'old' for Age) and created a set of nine 'IF-THEN' rules to predict diabetes risk. For instance, a rule might state: "IF BMI is high AND Age is old, THEN Diabetes Risk is high."
- **Results:** The model's output was visualized as a 3D surface plot. This plot clearly shows that as both BMI and Age increase, the predicted risk of diabetes also rises, which is a logical and expected outcome. The system's output is easily interpretable, making it a "white-box" model.

2.2 Multilayer Perceptron (MLP)

The MLP model, a type of artificial neural network, was implemented using `scikit-learn`. Unlike the fuzzy system, this model utilized all **eight clinical features** of the dataset.

- **Methodology:** The data was normalized and split into training, validation, and testing sets. We used a simple network with one hidden layer containing 10 neurons. The model was trained to perform a binary classification (diabetic vs. healthy).
- **Results:** The model's performance was evaluated using a **Confusion Matrix**. The matrix showed that out of the 231 test samples, the model correctly identified 123 healthy individuals (True Negatives) and 51 diabetic individuals (True Positives). However, it also made 28 false positive and 29 false negative predictions. This resulted in an overall accuracy of approximately **75.3%**.

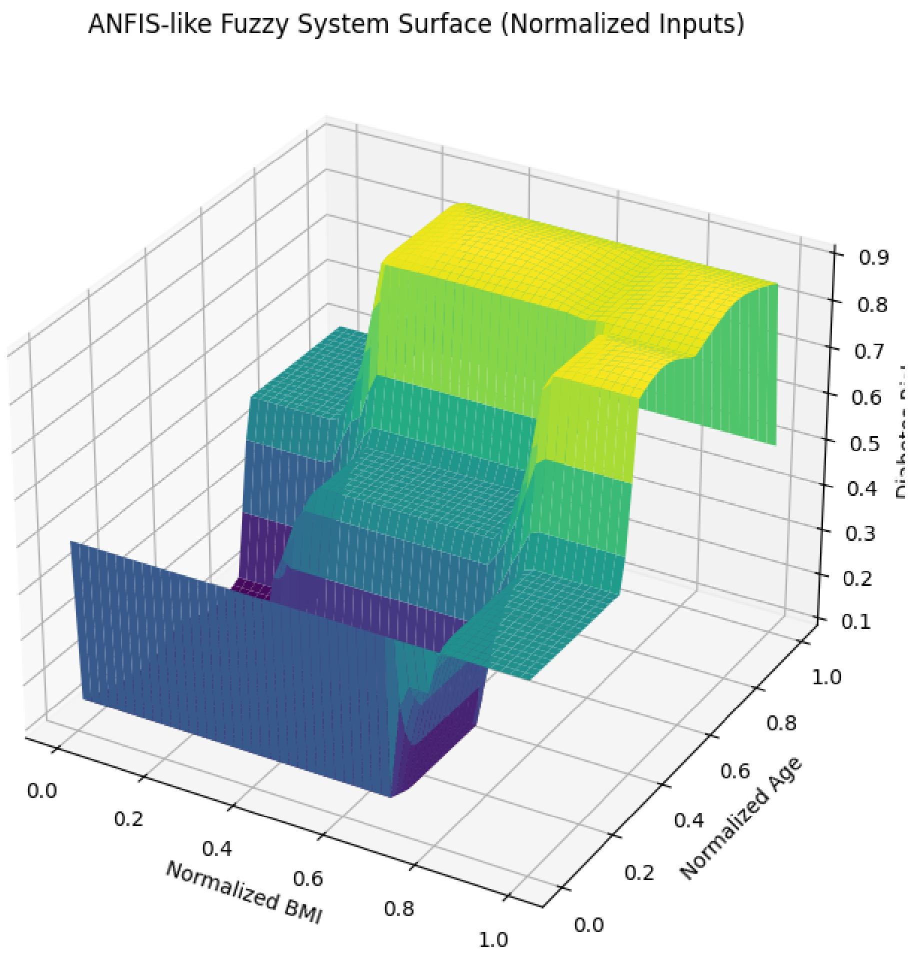

Figure 1: ANFIS-like Fuzzy System Surface (Normalized Inputs)

Criterion	**ANFIS Model (2 Inputs)**	**MLP Model (8 Inputs)**
Accuracy	Lower (due to fewer inputs)	Higher ($\approx 75.3\%$)
Interpretability	**Very High (White-box)**	**Very Low (Black-box)**
Strengths	Easy to understand rules; high explainability; simple structure.	Higher accuracy; uses all features; scalable.
Weaknesses	Lower accuracy; not efficient with high-dimensional data.	Lacks transparency; difficult to explain decision-making process.

Table 1: Comparison of ANFIS and MLP Models

3 Comparison and Conclusion

The MLP and ANFIS models were compared based on three key criteria:

The choice between these models for a medical application depends on the priority. While the **MLP** offers higher predictive accuracy by leveraging more data, it operates as a "black-box" that is difficult to explain. In contrast, the **ANFIS model**, despite its lower accuracy, is a "white-box" that provides clear, understandable rules. For assisting a doctor in diagnosing a patient, the ability to interpret and explain the model's reasoning is paramount. Therefore, the **ANFIS-like fuzzy system is arguably a more suitable choice for medical decision support**, as it provides a clear, logical basis for its predictions, fostering trust and understanding between the physician and the patient.

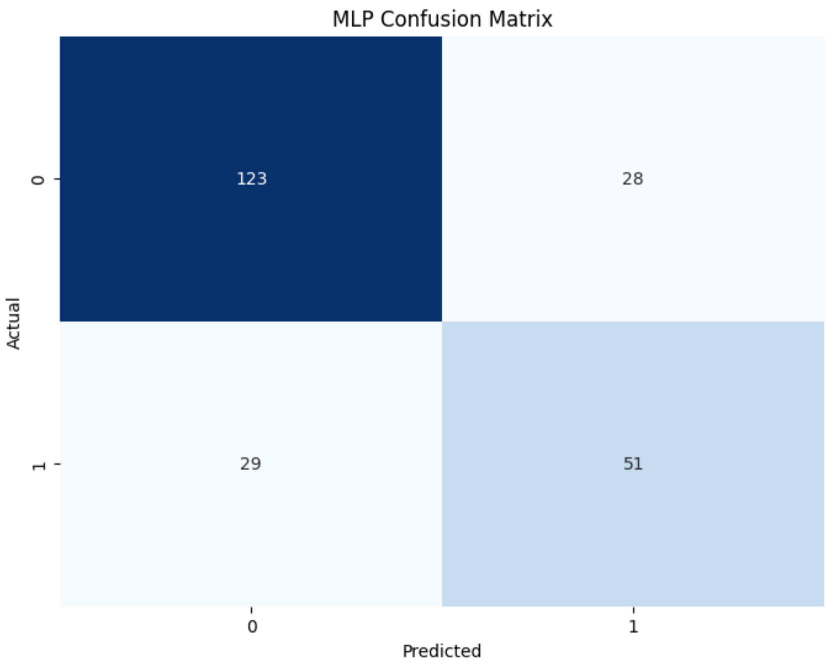

Figure 2: MLP Confusion Matrix

Project Link

- Google Colab Notebook